KOLHAPUR INSTITUTE OF TECHNOLOGY, KOLHAPUR

Department of Electronics and Telecommunications

IoT Based Incident Detection And Path Tracking System With GSM Alert

(Mega Project)

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CERTIFICATE

This is to certify that the Project report entitled, "**IoT Based Incident Detection And Path Tracking System with GSM Alert**" submitted by **Vaishnavi Patil (2223000427) Ballaleshwar Deshpande (2223000047), Anuradha Jadhav (2223000635)**, **Akash Meherkar (2223001017)**, in partial fulfillment of award of degree in B.Tech in **"ELECTRONICS AND TELECOMMUNICATION"** at KIT's college of Engineering, Kolhapur, Maharashtra, INDIA, is a record of work carried out under supervision and guidance.

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DECLARATION

I hereby declare that the Project entitled, **"*IoT Based Incident Detection And Path Tracking System With GSM Alert*"** submitted to KIT's College of Engineering, Kolhapur, Maharashtra, INDIA in the partial fulfilment of the award of the Degree of **"BAECHLOR OF TECHNOLOGY"** in **"ELECTRONICS AND TELECOMMUNICATION"** is a Bonafide work carried out by me. The material contained in this Project has not been submitted to any University or Institution for the award of any degree.

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INTRODUCTION

Accidental falls and disorientation are among the most significant challenges faced by elderly individuals and patients with limited mobility. Such incidents can lead to severe injuries or even life-threatening complications if timely medical assistance is not provided. Unfortunately, in many cases, the victim may be alone or unable to seek help immediately after the fall, resulting in delayed response and worsening health conditions. Existing safety solutions, such as smartwatches and advanced health monitoring systems, do offer fall detection and tracking features, but they are often expensive, require constant connectivity, or depend on smartphone compatibility — making them inaccessible for a large segment of the population, especially in rural and economically weaker regions.

To address this issue, this project proposes a low-cost, IoT-based Incident Detection and Path Tracking System with GSM Alert, designed to provide an affordable and reliable safety solution for individuals at risk. The system utilizes sensors such as an accelerometer and gyroscope to continuously monitor the user's movement and orientation. When an unexpected impact or fall is detected, it is identified as an incident, triggering the alert mechanism. The GPS module retrieves the user's live location coordinates, which are then automatically sent via GSM to a pre-registered caregiver or emergency contact through an SMS notification. This ensures immediate communication of the user's condition and location, allowing for faster rescue and medical response.

Beyond emergency alerts, the system also features path tracking capabilities that record the user's movement throughout the day. In case the user becomes disoriented or lost, the device provides navigation guidance to help retrace their steps back to a familiar or safe location. The compact and wearable design of the device ensures comfort and ease of use, making it suitable for daily wear by elderly people, patients recovering from surgery, and individuals with memory or mobility issues.

LITERATURE SURVEY

Accidental falls and disorientation among elderly individuals and patients with limited mobility are major causes of injury and loss of independence. To address this, researchers have developed intelligent fall detection and alert systems using Internet of Things (IoT) technologies to provide timely assistance and enhance safety.

Fall detection approaches are generally classified into three types: wearable sensor-based, vision-based, and smartphone-based systems. Among these, wearable sensor-based systems are preferred due to their portability, privacy, and real-time response. Studies have shown that combining accelerometers and gyroscopes effectively detects sudden motion changes that indicate a fall. Microcontrollers such as Arduino and communication modules like GSM or Wi-Fi further improve these systems by enabling automatic alerts to caregivers.

Newaz et al. (2023) reviewed existing fall detection techniques, noting that threshold-based algorithms are simple and affordable but may produce false detections. Zhang et al. (2024) proposed a deep-learning approach that improved accuracy but required high processing power, making it unsuitable for low-cost wearables. Tseng et al. (2025) developed a GPS-integrated wearable system to track users after a fall, but it faced limitations in indoor environments with weak GPS signals. Cullen et al. (2022) also highlighted the importance of GPS-based tracking for dementia patients to assist in cases of wandering or disorientation.

Prototypes using Arduino, GPS, and GSM modules have demonstrated the feasibility of affordable IoT-based fall detection systems. However, common issues such as power consumption, false alarms, and limited real-world testing remain challenges.

From the literature reviewed, it is evident that few systems combine incident detection, path tracking, and GSM-based alerting in one affordable solution.

PROBLEM STATEMENT

Elderly individuals and patients with limited mobility are highly prone to accidental falls and disorientation, which can lead to serious injuries, hospitalization, or even fatal outcomes if timely assistance is not provided. In many cases, victims are unable to seek help immediately, resulting in delayed medical response and worsening health conditions.

Although several smart devices such as health monitoring watches and advanced fall detection systems are available, they are often expensive, require constant internet connectivity, or depend on smartphones for operation, making them inaccessible to a large segment of the population especially in rural or economically weaker areas.

Furthermore, most existing systems focus only on fall detection or emergency alerting and lack additional functionalities such as path tracking and navigation assistance for users who experience disorientation. Power consumption, high device cost, and limited real-world testing remain major challenges in this field.

Hence, there is a need to develop a low-cost, IoT-based wearable system capable of detecting accidental incidents, tracking user movement, and sending real-time location alerts via GSM to caregivers. This project aims to address these limitations by providing a reliable, portable, and affordable safety solution for elderly individuals and patients with mobility issues.

OBJECTIVES

1. To design and develop an IoT-based wearable safety device capable of detecting accidental falls or sudden impacts using motion sensors such as an accelerometer and gyroscope.

2. To integrate GPS and GSM modules for real-time location tracking and automatic transmission of alert messages to caregivers or emergency contacts during an incident.

3. To implement a path-tracking feature that records the user's walking route and assists in retracing steps, reducing the risk of disorientation.

4. To develop a microcontroller-based control system that processes sensor data efficiently and ensures reliable detection and alert generation with minimal delay.

5. To ensure low power consumption and portability so that the device remains suitable for continuous, long-term use without frequent recharging.

6. To create an affordable and user-friendly solution that can be easily adopted by elderly individuals and patients in both rural and urban environments.

7. To enhance personal safety and healthcare monitoring by providing a practical system that bridges the gap between fall detection, path tracking, and emergency communication in a single integrated platform.

WORKING

The working of the proposed IoT-Based Incident Detection and Path Tracking System with GSM Alert is centered around continuous motion sensing, intelligent data processing, and automated communication. The system continuously collects movement data from the accelerometer and gyroscope, which detect variations in acceleration, angular velocity, and orientation of the user. When the sensors identify abnormal readings that exceed a predefined threshold, the microcontroller interprets this as a possible fall or accident. Once an incident is confirmed, the microcontroller immediately activates the GPS module, which determines the precise latitude and longitude of the user. These location details are then sent to the GSM module, which transmits an SMS alert containing both the incident message and live coordinates to a pre-registered emergency contact or caregiver, ensuring timely assistance and minimizing response delay.

Beyond accident detection, the system also provides path tracking functionality, where the GPS module records the user's walking path over time. This stored data can later assist in retracing steps if the user becomes disoriented or lost, offering an extra layer of safety and confidence to the user. The entire setup is powered by a rechargeable battery, making the device compact, lightweight, and portable for daily use. It can be worn comfortably on the body, allowing for continuous operation without restricting mobility.

All the modules — including the motion sensors, GPS, GSM, and microcontroller — work in close coordination to ensure efficient data collection, processing, and communication. The system has been designed to minimize power consumption while maintaining real-time performance, ensuring both accuracy and reliability. Overall, the proposed device provides an effective, low-cost, and user-friendly safety solution that combines incident detection, location tracking, and emergency alerting into one integrated IoT-based platform, significantly improving safety and independence for elderly individuals and patients with mobility challenges.

SIMULATION

The simulation of the proposed IoT-Based Incident Detection and Path Tracking System with GSM Alert was carried out using the Wokwi IoT simulation platform, which provides a virtual environment for testing embedded systems without the need for physical hardware. The main objective of this simulation was to verify the correct operation of sensors, communication modules, and the data transmission process before implementing the hardware prototype. The system uses an ESP32 microcontroller as the central processing unit, interfaced with an MPU6050 accelerometer and gyroscope, a SIM800L GSM module, and a NEO-6M GPS module.

In the simulation setup, the MPU6050 was connected to the ESP32 via the I2C communication protocol, with the SDA and SCL pins linked to GPIO21 and GPIO22 respectively. The sensor provided continuous acceleration and orientation data, which was processed by the microcontroller to detect any sudden impact or abnormal movement. The SIM800L GSM module was connected through UART communication (TX–RX configuration) and powered using a regulated 3.7 –4.2V supply. It was programmed to send SMS alerts to predefined contacts using AT commands whenever an incident was detected. The GPS module was also interfaced through UART, providing live latitude and longitude coordinates to track the user's position in real time.

During the simulation process, different motion conditions were created to test the threshold-based fall detection algorithm. When the system detected a sudden acceleration beyond the defined threshold, it identified the event as a fall or impact. The ESP32 then retrieved the user's current location through the GPS module and automatically transmitted the data to the GSM module, which successfully sent an SMS alert containing the user's live location. The message was observed on the virtual serial monitor in Wokwi, confirming the successful operation of the communication process. Additionally, the system continuously logged GPS data to simulate path tracking, verifying that the user's movement history could be recorded and later used to retrace steps in case of disorientation.

The simulation results proved that all modules worked efficiently in synchronization, and the system successfully achieved its objectives of detecting incidents, locating the user, and sending emergency alerts. The smooth communication between the ESP32, sensors, GPS, and GSM modules validated the feasibility of the design. The results also confirmed that the proposed system is technically sound, power-efficient, and ready for real-time implementation. Overall, the simulation demonstrated the effectiveness of the device in ensuring quick response, user safety, and reliability for elderly individuals and patients with mobility challenges.

COMPONENT COST

The IoT-Based Incident Detection and Path Tracking System with GSM Alert project was designed with the objective of developing a low-cost, efficient, and portable safety device. To ensure affordability and accessibility, all components used in the system were selected based on their availability, performance, and cost-effectiveness. The major hardware components include an ESP32 microcontroller, MPU6050 accelerometer and gyroscope sensor, SIM800L GSM module, NEO-6M GPS module, and a Li-ion battery for power supply. Supporting components such as the TP4056 charging module, buzzer, LEDs, resistors, and vibration motor were also incorporated to enhance the functionality and user interaction of the system.

The overall estimated cost of the components ranges between ₹2,000 to ₹2,500, depending on the market rates and availability. This makes the proposed design significantly more economical compared to commercial health monitoring or fall detection devices. The detailed component cost distribution is shown below:

Component Name	Price
ESP 32	400-600rs
MPU 6050	150-200rs
SIM 800L	400-600rs
GPS Module	400-600rs
Li-on Battery	250-400rs
TP4056 Charging Module	40-60rs
Slide Switch and Battery Connector	200rs
Buzzer , LED , Resistors , Vibration Motor	200rs

Total Estimated Cost: ₹2,000 – ₹2,500

This cost analysis confirms that the system provides a cost-effective and reliable alternative to existing commercial devices, maintaining both performance and affordability while ensuring easy deployment for elderly individuals and patients requiring safety monitor

CONCLUSION

The IoT-Based Incident Detection and Path Tracking System with GSM Alert provides a simple, low-cost, and efficient solution for ensuring the safety of elderly individuals and patients with mobility issues. The system accurately detects falls using motion sensors and immediately sends an SMS alert with GPS location through the GSM module. It also records the user's walking path to help retrace steps in case of disorientation. The simulation results confirm that all components work effectively, making the device reliable, portable, and suitable for real-time safety monitoring applications.

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